

'National Income' and 'Per Capita Income'

Growth and Development: Measurement of growth:

'National Income' and 'Per Capita Income'



What is Economics?

The subject matter of Economics focuses on **the rational management of scarce resources**. So that, these limited resources can be optimally transformed into valuable goods in order to maximise economic gains.

According to Adam Smith:

Subject matter of Economics: The subject matter of economics can be broadly classified into two groups:

Microeconomics

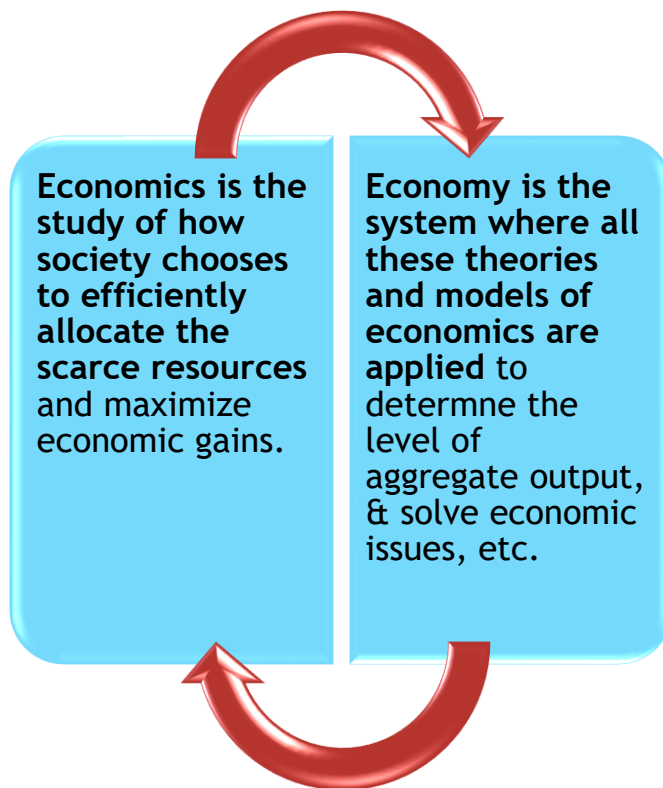
- When economic models are applied and studied onto an individual unit like a consumer, & a producer.
- E.g., The subject matter of microeconomics gave rise to the Consumer behaviour, Producer behaviour theories.

Macroeconomics

- When the economic issues and models are applied and studied at the aggregate level, it gives rise to the subject matter of Macroeconomics.
- Under this, economic issues like Inflation, National Income determination, Money supply, Unemployment and Poverty, etc. are studied.

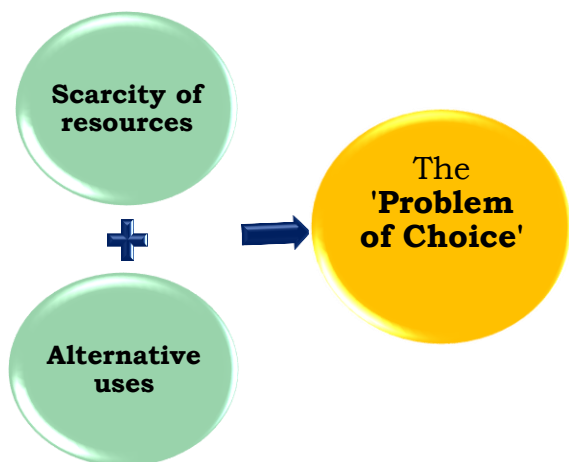
Economics vs Economy: For example, the consumer behaviour and producer behaviour theories, is used to determine the national output level.

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The CENTRAL PROBLEMS of an Economy:

Availability of resources does not necessarily translate into growth for an economy. The dearth of resources with a lot of alternative use poses a Problem of Choice. The producers and governments have to decide where to allocate these resources and which technology to use in the production processes.



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As a result, there are 'Three Central Problems' which every economy faces, irrespective of the amount and nature of resources available.



What to Produce?

Whether to produce more of capital goods or consumer goods?



How to Produce?

Whether to use labor-intensive or capital-intensive technique of production?



For whom to Produce?

Whether the allocation of the produce is promoting equality or not?

1. What to produce?

- Since **resources are limited**, every economy has to face this **trade-off of what kind of goods to produce.**
- **Consumer goods are necessary for survival** but **capital goods are a pre-requisite for industrial production.** Hence, the question is how much resources should be allocated to production of consumer goods (food, clothing, appliances, etc.) and capital goods (machinery, raw materials, etc.).

2. How to Produce?

- Technique of production is significant for optimum utilization of resources. But which technique should be adopted is an important choice to make.
- This dilemma can be best understood with the help of the following example:

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Populous countries can adopt a labor-intensive technique of production.



Countries with more of capital available easily in comparison to labor, can go for capital-intensive technique of production.

3. For whom to Produce?

This question deals with the distribution of the goods, income and proceeds from services rendered.

The distribution of resources should be such that it ensures:

- a) Reduction in inequality.
- b) Optimum allocation of the resources.
- c) Overall Social Well-being.

Types of Goods:

On the basis of 'Value Addition' and 'End Use', the goods produced in an economy can be broadly categorized into two groups:

- 1) **Final and intermediate goods.**
- 2) **Consumer and Capital goods.**

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❖ **On the basis of Value Addition:**

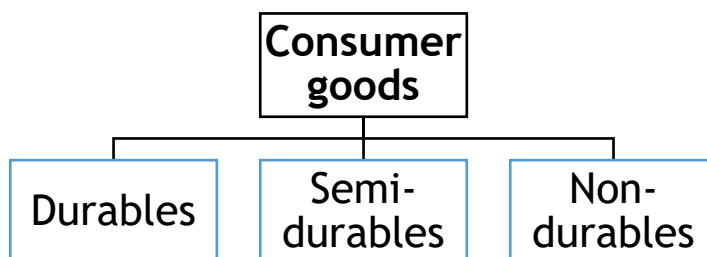
Intermediate Goods	Final Goods
Intermediate goods are those goods which are still in the process of value addition Or, these goods are to be used as raw material in some other industries. E.g., Wheat flour and Sugar are used as raw materials in a bakery.	Final goods are those which do not require further value-addition or which are ready for use. E.g., Books, clothes, shoes etc.

❖ **On the basis of their End-use:**

Consumer goods	Capital goods
These goods are consumed with the purpose of utility maximization. E.g., Biscuits, garments, shoes, etc.	Goods which are used in the process of production of goods and services or which aid in the process of wealth generation. E.g., machinery, equipments, building, etc.

- **NOTE:** The **same good can be a capital good and a consumer good**. It's **classification would depend upon the purpose of its use**. **E.g.,** an A. C. installed in a house is a consumer good, if the same A. C. gets installed in a factory, office or any such place where an economic activity is going on, it would be a capital good.

❖ **Consumer goods can further be classified into:**



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- **Durable goods:** Those goods, which can be used repeatedly for a long period of time as they have a long-life span like Automobiles, Mobile phones, Computers. etc.
- **Semi durable goods:** Those goods, which do not have very long-life span, but they do not get consumed in a single use as well. E. g. Footwear, crockery, clothing and other textiles, etc.
- **Non-durable goods:** These are the goods which have a very short life span and mostly get consumed in a single use. E. g. Fruits, milk, vegetable, dispo etc.

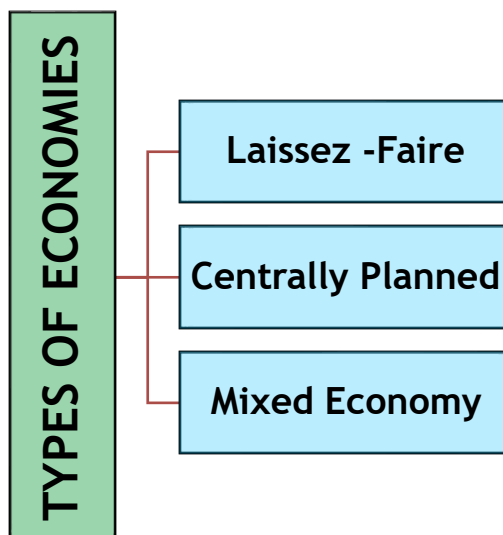


Note: 1) **Consumption Expenditure:** The expenditure incurred by an individual, household or the government, on final consumer goods. E.g., Food items purchased by an individual or household, stationery purchased by the government for its departments, etc.

2) **Investment Expenditure:** Expenditure incurred by a producer on the capital goods. E.g., purchase of machinery, factory equipments, etc.

Types of Economies and their approach towards the Central Problems:

- Every economy has its own arrangement for allocation of resources and distribution of the goods and services.
- But there are certain similarities on the basis of which the economies can be clubbed into **Three broad categories:**



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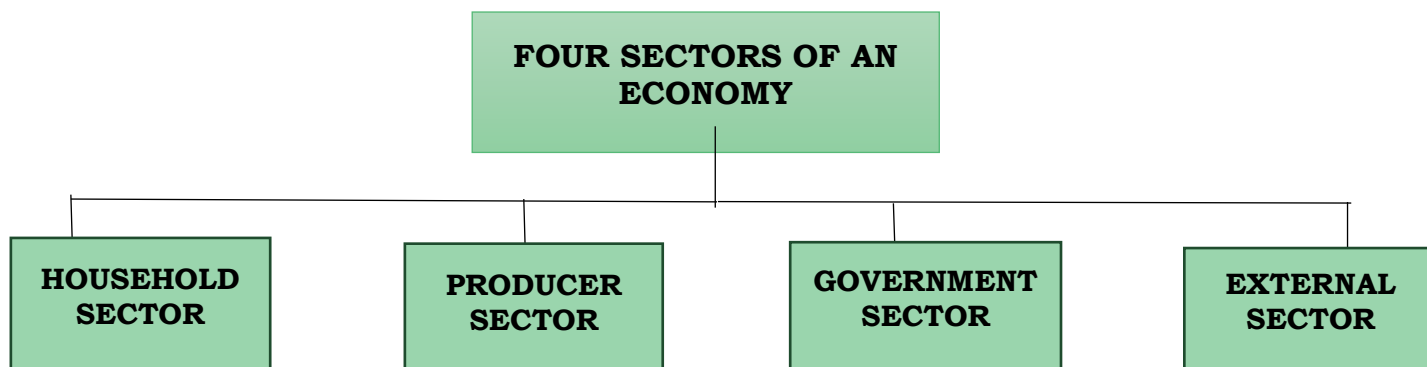
- ✓ **Laissez -Faire/Free Market Economy:** The free-market economy **finds solutions to the central problems** with the help of the market **forces of Demand and Supply**. It **does not mean absence of the government**, but the control of the state over the factors of production is negligible and profit motive dominates the process of production.
E.g., Singapore, Hong Kong, Netherlands, etc.
The producers in such an economy are free to decide, what to produce, how to produce and for whom to produce, as per the market demand and supply.
- ✓ **Centrally Planned Economy:** In such an economy, **the factors of production are majorly owned by the state**.
The motive of social welfare is given more priority over profit maximization. **E.g., China, Cuba etc.**
In such an economy, the decisions relating to the central problems are primarily taken by the state.
- ✓ **Mixed Economy:** A mixed economy has features of both, the free market and the planned economy. E.g., India.
Such economies have freedom of economic activities as well as government regulation up to certain extent to strike a balance between the aim of profit maximization and public welfare.
Industries of strategic importance like atomic energy, Space, Defence, etc. are not left completely left to the market forces of demand and supply. Such economies try to strike a balance between the aim of profit maximization and public welfare.



Note: Contemporary economies have features of both, the command as well as the mixed economy. It is difficult to classify them as exclusively capitalist or socialist. With globalization, almost every economy is highly linked with one another and the growth of Multinational Companies is making these boundaries blur for capitalism.

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- **Four sectors of an economy: An economy can be divided into following sectors:**



1. **The Household sector:** It refers to the consumers of goods and services. The households are also suppliers of the Factors of production as well.

Factors of Production: These are resources which are essential for production of any good or service. The income earned by these factors of production is called **Factor Income**. There are four factors of production:

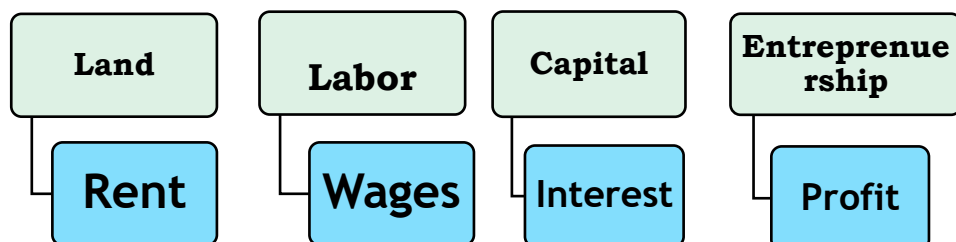
(a) **Land:** The return for land is **rent**.

(b) **Labor:** The return for labor is Wage, or **Compensation of employees**.

(c) **Capital:** Anything which is used in the process of production of any good or service is capital. It can be money, or any physical asset like machinery, human capital etc. **The return for capital is interest.**

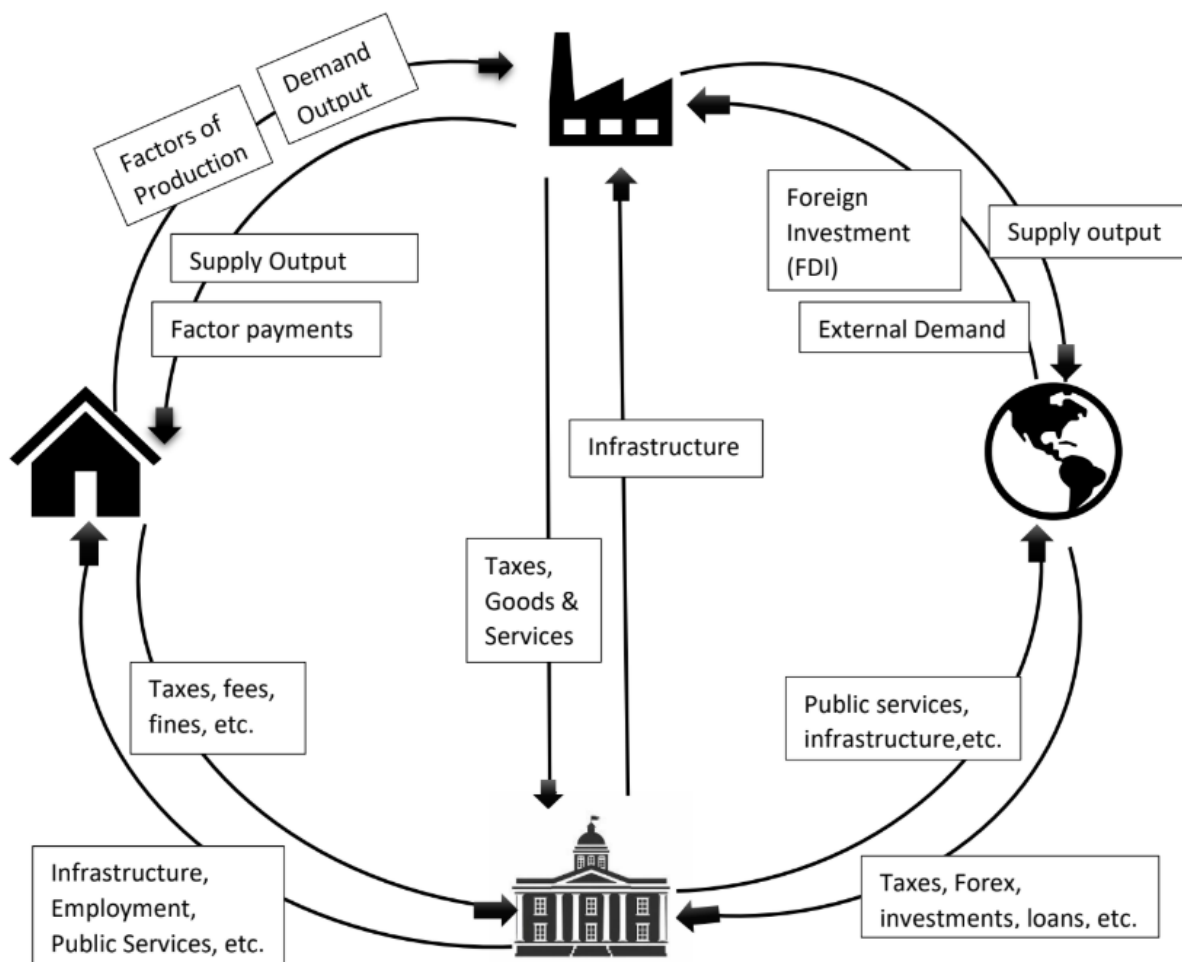
- **Entrepreneur:** An entrepreneur runs and manages the business/industry. **An entrepreneur earns profit.**

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- 2) **Producer Sector:** All the **producing units** in the economy can be clubbed into this sector.
 - 3) **Government sector:** The **government or the state** works as a **consumer**, a **producer** (NTPC Limited, Rural Electrification Corporation, etc.) and also ensure **public welfare**.
 - 4) **The External Sector/ Rest of the World:** All **trade related activities** (export and import) and **all the financial flows into and from the economy** (current account& capital account) constitute the external sector.
- **The Circular Flow of Income:** All these sectors of an economy **interact with each other with the help of Inputs and Outputs in the market**. The concept of **circular flow of money** or inter-sectoral flows, highlights the **direction of flow of economic activities** across these sectors i.e., the **Goods and services** (labor, land, etc) **generally flows in a clockwise direction** and the **factor payments or other money flows**, move in a **counter-clockwise direction**.

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➤ THEORY OF DEMAND

- **Demand:** It refers to **all the possible quantities of a commodity** that a **consumer is willing and able to buy** at **different possible prices** of that commodity, during a period of time.
- **Demand vs desire:** demand is the willingness to buy something along with the capacity to pay for it. On the other hand, desire is merely a wish to buy something, but the person does not have enough money to purchase that good.

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Quantity demanded: it refers to a specific quantity to be purchased at a specific price of a commodity.

➤ **Factors influencing Demand:**

The demand for any good or service depends upon so many factors. These factors are called as **determinants of demand**. They are:

- i. **Price of a good:** The demand for any commodity **firstly depends upon its own price**. When the price rises, the quantity demanded falls, and vice-versa.
- ii. **Income of the consumers:** An increase in the income of an individual, tends to increase the demand for a **Normal good**, at any given price. A decrease in income generally lowers the demand. This **happens when the good is a Normal Good**. But **when the good is an Inferior good**, an **increase in income will lead to a fall in the demand for the good** and a **decrease in income will lead to an increase in the demand for an inferior good**.



Normal goods- These are the goods for which, **the demand is directly related to consumer's income** i.e., with the rise in income, demand also rises and vice versa, e.g., Good quality clothing, good quality foodgrains etc.

Inferior goods- These are the goods for which the **demand is inversely related to consumer's income** i.e. with the **rise in income, demand falls and vice versa**, e.g., low quality foodgrains/groceries, public transportation, etc.

- iii. **Prices of substitute goods:** Substitute goods are the goods which **can be substituted for each other**, such as **tea and coffee** or **ball pen and ink pen**.

In the case of such goods, **increase in the price of one, causes increase in the demand for the other**(i. e. **direct relation between price of one good and the demand for the other one**).

- iv. **Price of complementary goods:** Complementary goods are those **goods which complete each other** and are therefore, demanded together in a fixed proportion. **E.g., Car and fuel, ink and pen, etc.** A **fall in price of one, causes an increase in the demand of the other and vice versa**.

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- v. **Taste and preferences of the consumers, population and weather:** Preference of a consumer changes according to fashion, weather conditions, climatic conditions etc. **E.g., Summers witness a spike in demand for ice-creams, any new model of a smartphone is demanded more, etc.**

➤ **Law of Demand**

The law states that, **ceteris paribus, quantity demanded of a commodity increases with a fall in its own price, and viced-versa.** There exists an **inverse relationship between price of a good and its quantity demanded.** Graphically, it is represented by a **downward sloping demand curve.**

➤ **Assumptions of the Law of the Demand**

- i. **Tastes and preferences** of the consumer remain **constant**
- ii. There is **no change in the income** of the consumers
- iii. **Prices of the related goods** do not change
- iv. **No change in the total assets**
- v. **No change in population**
- vi. **No expectation of further changes** in the availability of a commodity

➤ **Related terminologies:**

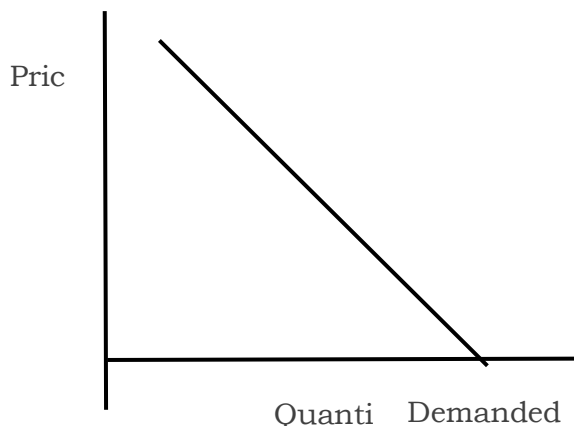
- **Individual demand-** Demand of a Good by an individual in the market.
- **Market demand-**It is the sum of individual demands of a commodity that all buyers are willing to buy at different prices.

➤ **Demand schedule-**Tabular representation of the law of demand:

Price of Apples in Rs.	Quantity demanded
100	4
150	3
200	2

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➤ Demand Curve:



➤ Exceptions to the Law of Demand:

The following are the exceptions to the Law of Demand.

- i. **Ostentatious goods:** These are purchased as a status symbol. E.g., **expensive cars, diamond jewellery**, etc. Such goods will be purchased more at a higher price and less at a lower price.
- ii. **Giffen goods**-This is observed by **British Economist, Sir Robert Giffen**. It is a **special type of inferior good whose income effect is negative, but price effect is positive. i. e. demand increases with rise in price and vice versa**. Giffen goods are the **basic necessity, typically food, for the nation, consumers spend significant portion of their income** on this good. There are **no close substitutes** for the good. E.g., **rice, salt, potatoes, and bread**.
- iii. **Shares or speculative market**-In the speculative market, particularly in the stock market, **more will be demanded when the prices are rising** and vice versa.
- iv. **Bandwagon effect:** When the consumers 'choice for goods and services is highly influenced by society's taste, or fashion trends. E.g., **Carrying an i-phone** is more of a status symbol than a necessity.
- v. **Veblen effect**-The belief in the false idea that high priced goods will have a better quality as compared to the low price ones.

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➤ THEORY OF SUPPLY:

There is a **difference between stock of the goods and supply of goods.**

Supply means part of stock of the goods which a **seller is willing to and able to sell at a particular price in the market, during a particular period of time.**

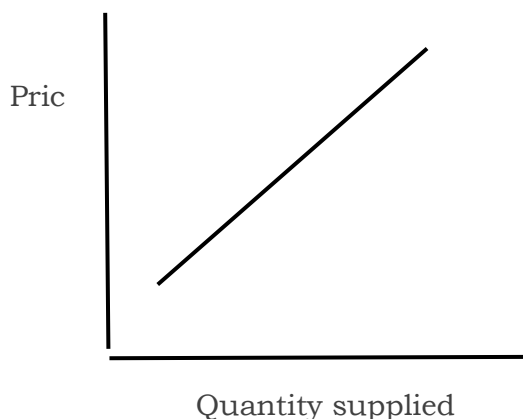
Law of supply- Other things remaining constant, price of a good and its quantity supplied are positively related to each other. Graphically, it is represented by an upward sloping supply curve.

➤ Assumptions of the law of supply:

- a) There is **no change in the price of factors of production**
- b) There is **no change in the techniques of production**
- c) There is **no change in the goal of the firm**
- d) There is **no change in the price of related goods**
- e) **Investors have full confidence over business**

Supply schedule: Tabular representation of the Law of Supply.

✚ **Supply Curve:**



Investment

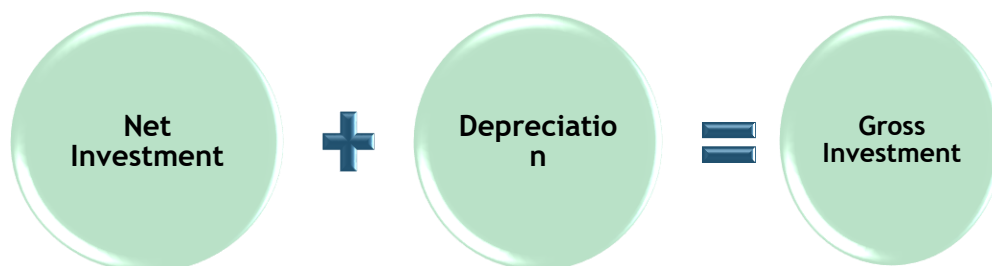
What is Investment?

Investment is a process **which creates assets or capital.** Or, it can be understood as that part of the GDP which is further used in the process of production.

Investment is a flow concept and **capital is a stock concept**. Since, investment is the process which creates capital, so one can define investment as a **'A flow which adds to the stock of capital.'**

+ Gross vs Net Investment:

- **Gross Investment:** Gross investment **refers to the purchase of fixed capital** (machinery, equipments, etc.) along with the expenditure on purchase of inventory stock. **Gross investment includes the expenditure incurred for replacement of the depreciated capital.**
- **Net Investment:** When the **replacement cost (expenditure incurred for replacement of the worn-out assets)** is **excluded from gross investment**, it gives Net Investment.



- **Stock:** A quantity measured at a point of time is a Stock. **E.g., Deposits with banks, stock of capital assets, etc.**
- **Flow:** A quantity measured during a period of time. **E.G., income, capital formation.**
- **Inventory Stock/Inventory Investment:** It refers to the stock of all such goods which have not been sold, plus the intermediated goods, and the raw materials.
- **Depreciation:** The fall in an assets value over the period of time. It is also known as, **'Consumption Of Fixed Capital'**. **E.g., wear and tear in machinery** leads to fall in its economic value over the period of time.

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National Income

National Income accounting is one of the primary concerns for an economy. It is an **important indicator of economic growth and employment**.

Gross Domestic Product: GDP is the key concept in national income accounting.

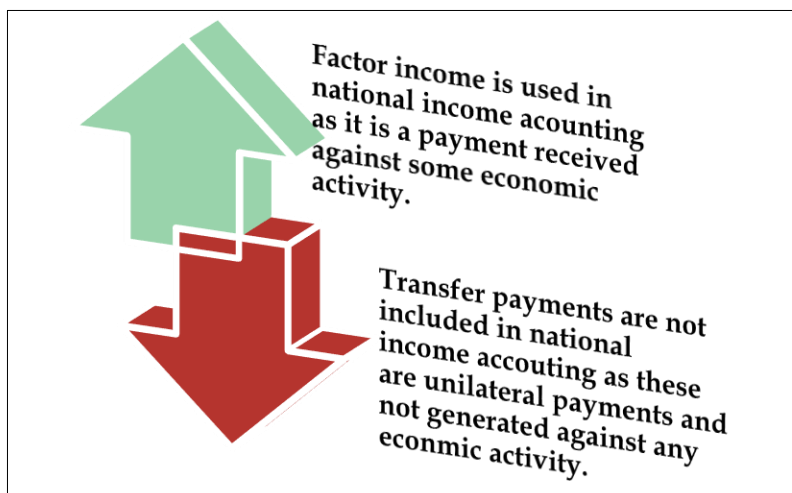
GDP growth is important because, it reflects the level of production and consumption, which in turn, determines the employment opportunities, and the overall growth of an economy.

GDP refers to the total **market value** of all **final goods and services** produced by **factors of production** within the **domestic territory** of a country within an accounting year.

Factor income: Income earned by the factors of production. Or, it can be defined as the payment received against an economic activity. **E.g., wages, rent, profit, interest.**

Transfer income: Transfer payments are **unilateral payments**. These are **not used in national income calculation** because they are not earned against any economic activity. **E.g., Unemployment allowance, grants, donations, old age pension, etc.**

Note: Only Factor Income is taken into consideration in national income accounting.

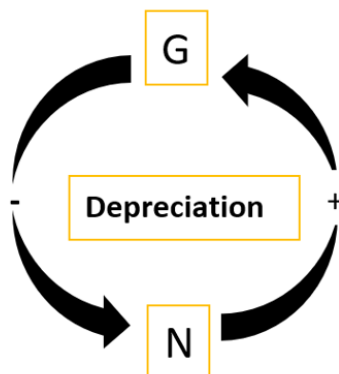


Domestic Territory: The domestic territory is not only the geographical area under jurisdiction, it also includes:

- Aircrafts and shipping vessels operated by our residents.
- Military bases, Embassies, Consulates of India in abroad.
- Other offshore platforms, boats, vessels, etc.

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Gross Domestic Product vs Net Domestic Product: Domestic product or income is generally measured as the Gross Domestic Product and Net Domestic Product.



- **Gross Domestic Product** is the total market value of all final goods and services, produced within the domestic territory, by factors of production, during an accounting year, **including depreciation**.
- **Net Domestic Product:** It is the total market value of all final goods and services, produced within the domestic territory, by factors of production, during an accounting year, **excluding depreciation**.

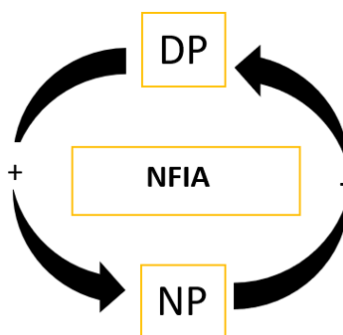
Example:

- **GDP** = Rs. 1000 cr
- **Depreciation** = Rs. 50 cr
- **NDP** = Rs. 950 cr

Domestic vs National Product: **Domestic Product** is the total market value of all final goods and services produced during an accounting year, within the domestic territory, **irrespective of who is producing it, normal residents or non-residents. It does not include the Net Factor Income from Abroad.**

- **National Product** is the **sum total of all the factor incomes earned by the normal residents of a country**, irrespective of where it has been generated, within the domestic territory or outside it. **It includes the Net Factor Income from Abroad.**

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Case study: Consider a Honda Plant in U.S. It is owned by Ohio Corporation (Japanese company), but most of the workers there are from U.S.

The output of the plant is calculated in U.S. GDP but, only a part of it is included in U.S. GNP. The wages paid to the U.S. workers will be added in its GNP. But, the profits from the plant will not be a part of US GNP. The profits will be counted in Japanese GNP as the plant is owned by Japanese entrepreneur so profits will go to them. The profits will be counted in Japan's GNP (NOT IN GDP as the production did not happen in Japan).

Source: Case and Fair, Principles of Economics, 9th Edition.



Residents: A taxpayer would qualify as a resident of India if he satisfies one of the following 2 conditions:

1. Stay in India for a year is 182 days or more in previous year or,
2. Stay in India for the immediately 4 preceding years is 365 days or more and 60 days or more in the relevant financial year.

Note: The **national product** measures all goods and services arising out of economic activity while **national income** is the sum of all incomes as a result of the economic activity. **These two are synonymous.**

- Since the production of goods and services is the result of the use of the factors of production, the process automatically generates income. **This income is in the form of return to the factors of production. Hence, these words are used interchangeably.**
- **Market Price vs Factor Cost:** Domestic Product or National Product can be measured at market price or factor cost. **Market price includes the Net Indirect Taxes**, while Factor cost does not.

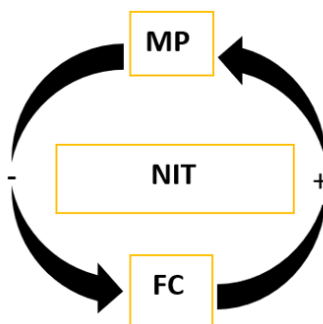
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$$\text{Net Indirect Taxes} = \text{Indirect Taxes} - \text{Subsidies}$$

GDP_{MP}: **Total market value** of all final goods and services produced within the domestic territory, by factors of production, during an accounting year.

GDP_{FC}: When the domestic product is calculated as **sum total of all the factor incomes** (not as the total market value), generated within the domestic territory of a nation, during an accounting year, it gives GDP at factor cost.

$$\begin{aligned} \text{GDP}_{\text{MP}} - \text{Net Indirect Taxes (NIT)} &= \text{GDP}_{\text{FC}} \\ \text{GDP}_{\text{FC}} + \text{NIT} &= \text{GDP}_{\text{MP}} \end{aligned}$$



➤ National Income and related aggregates:

GDP_{MP}: Gross Domestic Product at Market Price: **Total market value** of all final goods and services produced within the domestic territory during an accounting year, including depreciation.

GDP_{FC}: When GDP is calculated after **summing up the factor incomes (wages, rent, interest, profit)** generated within the domestic territory of a country during a financial year.

$$\begin{aligned} \text{GDP}_{\text{MP}} - \text{Net Indirect Taxes} &= \text{GDP}_{\text{FC}} \\ \text{GDP}_{\text{FC}} + \text{Net Indirect Taxes} &= \text{GDP}_{\text{MP}} \end{aligned}$$

GDP: Total market value of all final goods and services produced within the domestic territory of a country during an accounting year, **including depreciation**.

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NDP: Total market value of all final goods and services produced within the domestic territory of a country during an accounting year, **excluding depreciation.**

$$\text{GDP} - \text{Depreciation} = \text{NDP}$$

○ **GNP:** $\text{GDP} + \text{Net Factor Income from Abroad}$

○ **NNP:** $\text{GDP} + \text{NFIA} - \text{Depreciation}$

○ **GDP_{MP}** = $\text{GDP}_{\text{FC}} + \text{Net Indirect taxes}$

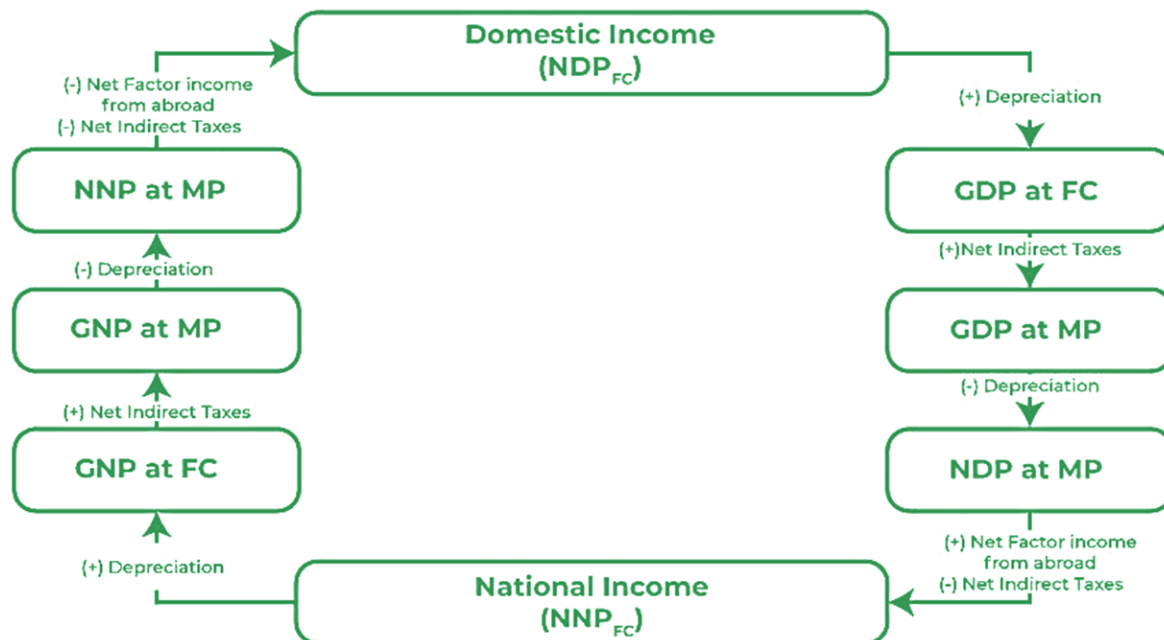
○ **GDP_{FC}** = $\text{GDP}_{\text{MP}} - \text{Net Indirect Taxes}$

GVA at Basic Prices: In the revision of National Accounts statistics done by Central Statistical Organization (CSO) in January 2015, it was decided that sector-wise estimates of Gross Value Added (GVA) will now be given at basic prices instead of factor cost.

$$\begin{aligned} \text{GVA at factor cost} + (\text{Production taxes less Production subsidies}) &= \text{GVA at basic prices} \\ \text{GDP at market prices} &= \text{GVA at basic prices} + \text{Product taxes} - \text{Product subsidies} \end{aligned}$$

- **Basic prices exclude any taxes on products the producer receives from the purchaser and passes on to government but include any subsidies the producer receives from government and uses to lower the prices charged to purchasers.**
- In simple terms, for any commodity the **basic price is the amount receivable by the producer from the purchaser for a unit of a product minus any tax on the product plus any subsidy on the product.**
- **However, GVA at basic prices will include production taxes and exclude production subsidies available on the commodity.**
- **On the other hand, GVA at factor cost includes no taxes and excludes no subsidies and GDP at market prices include both production and product taxes and excludes both production and product subsidies.**

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Exclusions in calculation of GDP: Such transactions in which, the goods, services or money are not being created afresh. **E.g., the second-hand cars or any other second-hand good** will not be considered for GDP calculation.

Calculating the transactions involving second hand or used items can lead to the problem of **double counting**.

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MISCELLANEOUS:

The production identities

Gross value added = Output – Intermediate consumption

GDP at Market Prices = Gross value added + Indirect taxes – Subsidies

GDP = final consumption expenditures/actual final consumption + changes in inventories + gross fixed capital formation + acquisition less disposals of valuables + exports of goods and services – imports of goods and services

GDP at Market Prices = compensation of employees + operating surplus / mixed income + CFC+ Indirect taxes – Subsidies

GDP at factor cost = GDP at market prices – indirect taxes + subsidies

NDP = GDP – Consumption of fixed capital

Income and saving identities

Gross National Income (GNI) = GDP at market prices + taxes less subsidies on production and imports at market prices (Net receivable from abroad) + Compensation of Employees (Net receivable from abroad) + property income (Net receivable from abroad)

Net National Income (NNI) at market prices = GNI at market prices – consumption of fixed capital

Net National Disposable Income (NNDI) = NNI + net taxes on income and wealth receivable from abroad + net social contributions and benefits receivable from abroad.

Net saving = NNDI – final consumption expenditure + net equity of households on pension funds receivable from abroad + net capital transfers receivable

Net saving + net capital transfers = changes in net worth due to savings and capital transfers

Saving and investment identities

Net saving + net capital transfers receivable = gross fixed capital formation – CFC + changes in inventories + acquisitions less disposals of valuables and non-produced non-financial assets + net lending/ net borrowing

Net lending (+)/borrowing(–) = net acquisitions of financial assets less net incurrence of financial liabilities

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✚ Formulae associated with all the National Income Aggregates:

1. **GVA at basic prices (Production Approach)** = Output at basic price – Intermediate Consumption
2. **GVA at basic prices (Income Approach)** = CE + OS/MI + CFC + Production taxes less Production subsidies
3. **GDP** = $\sum$ GVA at basic prices + Product taxes less Product subsidies(ii)
4. **NDP/NNI** = GDP/GNI - CFC
5. **GNI** = GDP + Net primary income from ROW (Receipts less payments)
6. **Primary Incomes** = CE + Property and Entrepreneurial Income
7. **NNDI** = NNI + other current transfers(iii) from ROW, net (Receipts less payments)
8. **GNDI** = NNDI + CFC = GNI + other current transfers(iii) from ROW, net (Receipts less payments)
9. **Gross Capital Formation (Financing Side)** = GrosEXs Savings + Net Capital Inflow from ROW
10. **GCF (Expenditure Side)** = GFCF + CIS + Valuables
11. **Gross Disposable Income of Govt.** = GFCE + Gross Saving of General Government
12. **Gross Disposable Income (GDI) of Households** = GNDI – GDI of Govt. – Gross Savings of All Corporations

REMARKS ON THE FORMULAE

- (i) *Production taxes or subsidies are paid or received with relation to production and are independent of the volume of actual production. Some examples are:*

Production Taxes - Land Revenues, Stamps & Registration fees and Tax on profession

Production Subsidies - Subsidies to Railways, Subsidies to village and small industries.

- (ii) *Product taxes or subsidies are paid or received on per unit of product. Some examples are:*

Product Taxes - Excise Duties, Sales Tax, Income Tax, Corporation Tax, etc.

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- (iii) *Other Current Transfers refers to current transfers other than the primary incomes.*
- (iv) *Gross Capital Formation (GCF) at the current as well as the constant prices is estimated by two approaches: – (i) through flow of funds, derived as Gross Saving plus net capital flow from Rest of the World (RoW); and (ii) by the commodity flow approach, derived by the type of assets. The estimates of GCF through the flow of funds approach are treated as the firmer estimates. GCF by industry of use and by institutional sectors does not include ‘valuables’ and therefore, these estimates are lower than the estimates available from commodity flow approach.*

Methods of calculating National Income

The **Circular Flow of Money Model** highlights **three sides of an economic activity:**

The production, The expenditure and The Income generation. All these phases of production are highly interlinked to each other. Therefore, National income can be calculated in three different ways:



- 1) The expenditure approach
- 2) The Value-Added method / The Product Method
- 3) The Income method

'National Income' and 'Per Capita Income'

The Product Method/ Value Added Method: It measures the National income as value addition by all the producing units or different sectors in an economy, in an accounting year. Or, by **summing up the value added at each stage of production in an economy.**

Gross Value added: Gross Value Added (GVA) is a metric that **measures the value that producers have added to the goods and services they have bought.**

$$\text{Value Added} = \text{Value of Output} - \text{Intermediate Consumption}$$

Producing firms	Value of Output	Intermediate Consumption	Value Added
Farmer	600	200	400
Flour mill	800	600	200
Bakery	1000	800	200
Shopkeeper	1200	1000	200
Total	3600	2600	1000 (GVA)

Source: T.R Jain & V.K. Ohri, Introductory Macroeconomics.

Difference between GDP and GVA?

GDP gives **demand side view of the output.** It reflects the demand of the goods and services produced within the domestic territory.

GVA reflects the supply side of the economy. It shows the sectoral contribution of each sector in overall GDP like, how much the agricultural, industrial, services sectors have contributed to the overall GDP.

It also helps the government identify sectoral deficiencies and formulate policies accordingly.

GVA: Gross Value Added: It shows the **sectoral contribution in the overall economic growth.** GDP is more of demand side indicator of economic growth and GVA is the supply side or producer's side indicator. **It is helpful in policy formulation as it gives the data related to sectoral performance. The government can find out the sectoral deficiencies and can formulate policies accordingly.**

'National Income' and 'Per Capita Income'

✚ Sectors of Indian Economy:

- **Primary sector:** When a good is produced by exploiting natural resources, it is an activity of the primary sector. Since most of the natural products we get are from agriculture, dairy, fishing, forestry, **this sector is also called agriculture and related sector.**
- **Secondary sector:** It covers activities in which **natural products are changed into other forms** through ways of manufacturing. Since this sector gradually became associated with the different kinds of industries that came up, it is **also called as industrial sector.**
- **Tertiary sector:** These are activities that help in the development of the primary and secondary sectors. These activities, by themselves, do not produce a good but they are an aid or a support for the production process.

Transport, storage, communication, banking, trade are some examples of tertiary activities. **Since these activities generate services rather than goods, the tertiary sector is also called the service sector.**

✚ The services sector can be divided into:

- **Quaternary Sector:** Quaternary activities involve the collection, production and dissemination of information or even the production of information. **Quaternary activities centre around research, development** and may be seen as an advanced form of services involving **specialised knowledge and technical skills. E.g., mutual fund managers, tax consultants, software developers, etc.**
- **Quinary Sector:** The highest level of decision makers or policy makers perform quinary activities.

Quinary activities are services that **focus on the creation, re-arrangement and interpretation of new and existing ideas**; data interpretation and the use and evaluation of new technologies. Often referred to as **'gold collar'** professions, they represent another subdivision of the tertiary sector representing special and highly paid skills **of senior business executives, government officials, research scientists, financial and legal consultants, etc.**

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Classification of the economic activities on the basis of rules and regulations that are followed as regards their employment:

- **Organised sector:** covers those enterprises or places of work where the **terms of employment are regular** and therefore, people have assured work. They are **registered by the government and have to follow its rules** and regulations which are given in various laws such as the Factories Act, Minimum Wages Act, Payment of Gratuity Act, Shops and Establishments Act etc.

Workers in the organised sector **enjoy security of employment.**

They get paid leave, payment during holidays, provident fund, gratuity etc. **They are supposed to get medical benefits**, pension etc.

- **Unorganised Sector:** The unorganized workers are paid a very low salary. They are often exploited and not paid a fair wage. Their earnings are low and not regular. These jobs are not secure and have no other benefits.

✚ The Expenditure Method:

Under this method, the National Income is calculated as the **sum of all the expenditures incurred on the purchase of final goods and services** produced in an economy during an accounting year.

Note: The expenditure is final.

The expenditures can be classified as:

C: Private Final Consumption Expenditure: Household spending on the final consumer goods.

I: Investment Expenditure: The Investment done by Firms and Households on capital, machinery, equipments, inventory, etc.

G: Government Final Consumption Expenditure: Government spending on Public goods and services.

Net Exports(X-M): Exports- Imports

$$Y = C + I + G + (X - M)$$

The expenditure method gives us the **GDP at market price.**

$$GDP_{MP} - \text{Depreciation} + NFIA - NIT = NNP_{FC}$$

'National Income' and 'Per Capita Income'

Example:

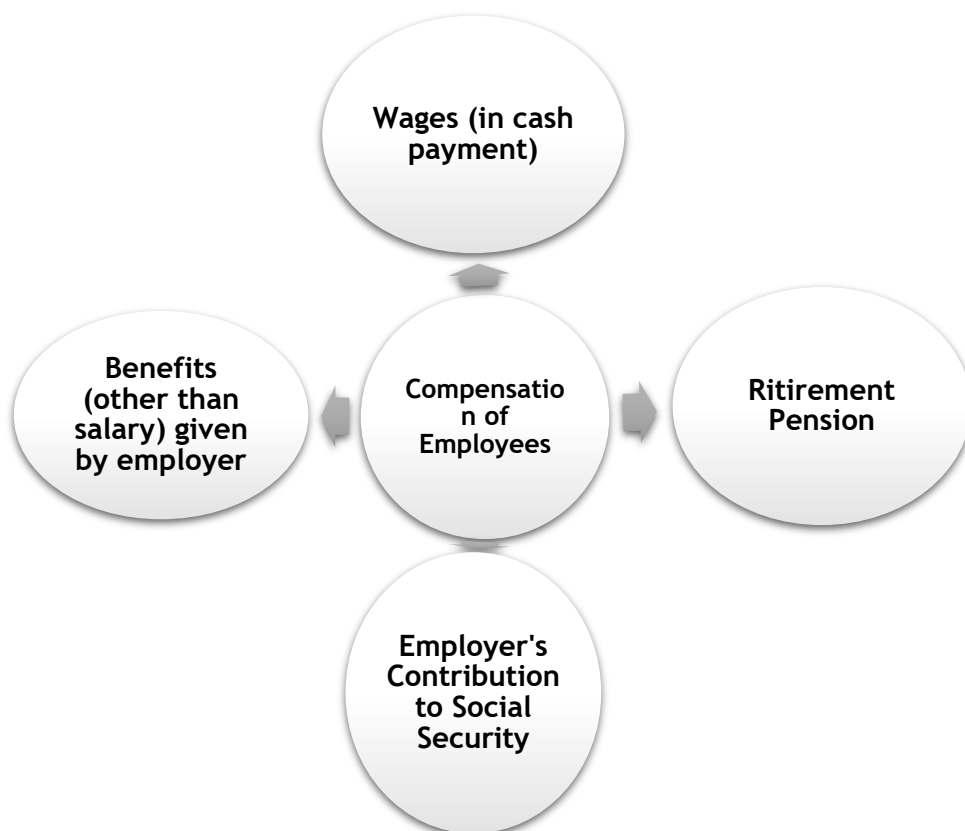
- C = 50, 000
- I = 30, 000
- G = 40, 000
- Exports = 70, 000
- Imports = 90, 000
- GDP_{MP} = 1, 00, 000

The Income Method:

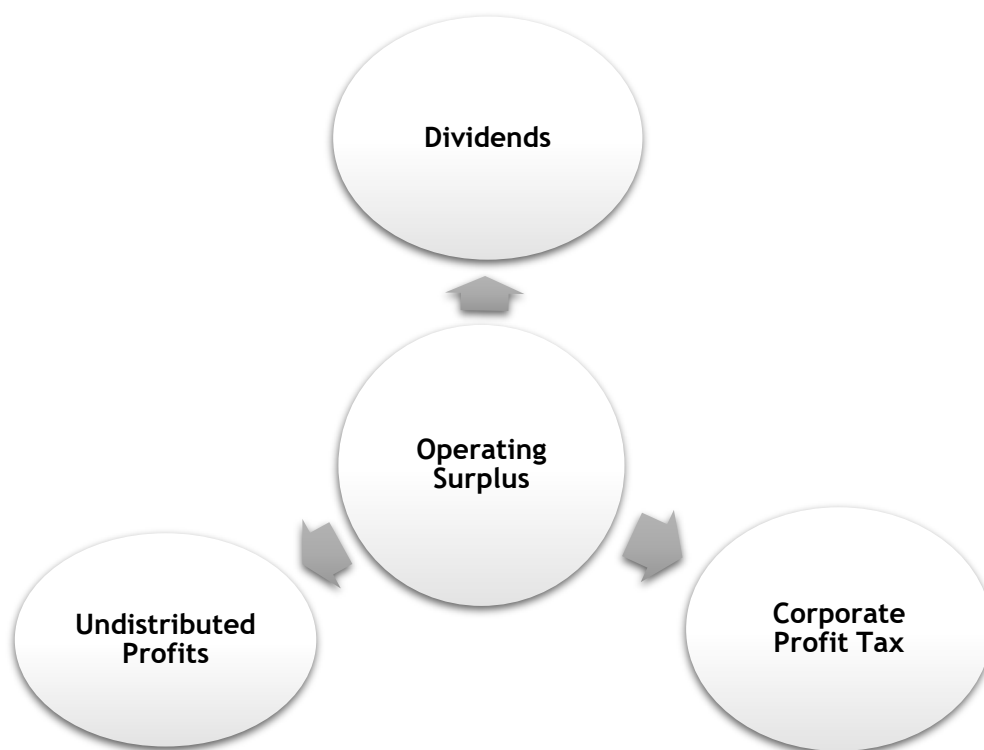
Under this method, **national income is calculated by way of factor payments received by the factors of production**, within the domestic territory, during an accounting year.

For the purpose of calculation of national income by Income Method, the **incomes can be categorized as following:**

$$\text{NDP}_{\text{FC}} = \text{Compensation of Employees} + \text{Operating Surplus} +$$



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Mixed Income: The **income of the self-employed persons** who are using their own factors of production like land, labor, capital, and entrepreneurship.

✚ **Other related concepts:**

- **Gross Capital Formation:** Acquisition of **fixed assets** (buildings, machinery, etc.) and **accumulation of stocks (change in stock of raw materials, semi-finished goods, etc.)** Or, it can be defined as the part of country's total expenditure which is not consumed but, added to nation's fixed assets and stocks.
- **Savings:** It reflects **excess of current income over current expenditure across various sectors of any economy**. In a closed economy, savings can be referred to as total capital formation, while in an open economy, savings is equal to total capital formation plus the net capital inflow from abroad.
- **National Disposable Income:** It is the sum total of **income from all the sources** available for a country's disposal or saving, during an accounting year.

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Net National Disposable Income = NNI + net taxes on income and wealth receivable from abroad + net social contributions and benefits receivable from abroad.
(NNDI)

Note: It includes factor incomes as well as the transfer payments.

National Disposable Income can be further classified as:

- 1) **Gross National Disposable Income:** GNDI is inclusive of depreciation.
 - 2) **Net National Disposable Income:** NNDI excludes depreciation.
- **Private Income:** It excluded all forms of income earned by government departments or entities. **Private Income is a measure of income derived from national income, by adding the sum of govt transfer payments and interest on national debt, and subtracting profits and property income of government entities/departments.**
 - **Personal Income:** It is the sum total of all the income actually received by individuals and households, from all sources (both factor income and transfer payments).

Personal Income = Private Income – UDP (Undistributed Profits retained by corporates) – Corporate Profit Tax.

- **Personal Disposable Income:** The part of personal income which, an individual, or a household can use at their will, after paying all the taxes, fees, fines, etc.

Personal Disposable Income = Personal Income – Direct Taxes – (Fees, Fines etc.)

Per Capita Income: National Income OR Total Income / Total Population.

Nominal and Real GDP:

GDP can be calculated at both, constant prices and current prices.

Nominal GDP: When GDP is calculated at Current prices. These prices are the ongoing prices in the accounting year.

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Nominal GDP = $P \times Q$

P = Current year prices

Q = Final goods and services produced during the accounting year

- **Real GDP:** It is calculated at the **base year prices**. Presently, the base year is **2011-12**.

Note: **Government is planning to change the base year** of national income calculation from **2011-12 to 2022-23**, by February 2026.

Real GDP = $P \times Q$

P = Base year prices

Q = Final goods and services produced in the accounting year

NOTE: Real GDP is more appropriate indicator of economic growth in comparison to the Nominal GDP. Because, Real GDP will only increase if the Quantity produced during the accounting year increases (Price remaining constant).

While nominal GDP might increase due to rise in the price level.

- **GDP DEFLATOR:** It measures the ratio of nominal to real GDP and shows the change in price level from base year to current year. **It is also known as Implicit Price Deflator.**

$$\frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Example: Nominal GDP = \$12 bn

Real GDP: \$10 bn

Deflator= 120 (it shows, from the base year, 20% rise in price level is there).

Limitations/Challenges of GDP

Although GDP is a good indicator of economic growth, it may not be sufficient as an indicator of economic development.

1) Domestic or non-market activities like housework or child care, are not counted in GDP calculation. If the same work is being done by a household help, it will contribute to GDP.

2) GDP does not reflect social ills. For example, a country might be witnessing an increase in GDP numbers due to an increase in production of weapons or munition, etc.

Increase in GDP might result into a environmental pollution to a great extent, which is not taken into consideration for national income accounting.

3) Distribution of the output: GDP only reflects the amount of output produced within the domestic territory. It does not reflect the distribution of the output or income proceeds.

If most of the output produced goes in the hands of a few people, then **GDP cannot be used to measure social- welfare.**

4) The problem of parallel economy and unorganized/informal economy: GDP does not take into account the transactions involving black money, or the unorganized sector's transactions. The parallel economy plays an important role in India's Economic growth. But, GDP only takes into account the organized or formal sector's income.

Tax evasion is an incentive for people to participate in the parallel/underground economy.

5) Externalities: Externalities refer to either the positive or negative consequences of an activity on its surroundings. **Negative externality:** a factory producing goods is added into GDP but the polluted waste discharged into the nearby river is not taken into consideration, which might be causing harm to the riverine life and people located nearby. Therefore, GDP as a measure of social welfare may be overestimating in nature.

Positive externality: GDP might also underestimate the positive impact on social welfare. E.g., construction of parks, urban greens etc.

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Green GDP

The green GDP is the measurement of GDP growth with the environmental consequences of that growth factored in. Green GDP accounts for the monetized loss of biodiversity, costs caused by climate change etc. **It is a measure of how a country is prepared for sustainable development.**

In 1987, the United Nations Brundtland Commission defined sustainability as
“meeting the needs of the present without compromising the ability of future generations to meet their own needs.”

As far back as in 2009, the Centre had announced its intention to unveil “green GDP” figures that account for the environmental costs of depletion and degradation of natural resources into the country’s economic growth figures. Subsequently, the Ministry of Statistics and Programme Implementation **set up an expert group in 2011 led by Partha Dasgupta**, to work out a framework for green national accounts in India, which submitted its report in March 2013. The process, is yet to be completed.

WORLD BANK AND IMF COUNTRY CLASSIFICATION

The World Bank Group assigns the world’s economies into four income groups: low, lower-middle, upper-middle, and high.

The classifications are updated each year on July 1, based on the GNI per capita of the previous calendar year. **GNI measures are expressed in United States dollars using conversion factors derived according to the Atlas method.** The World Bank’s income classification aims to reflect a country’s level of development, drawing on *Atlas GNI per capita* as a broadly available indicator of economic capacity.

The latest classification is as mentioned below:

As per this classification, India **comes under the Lower- middle income category.**

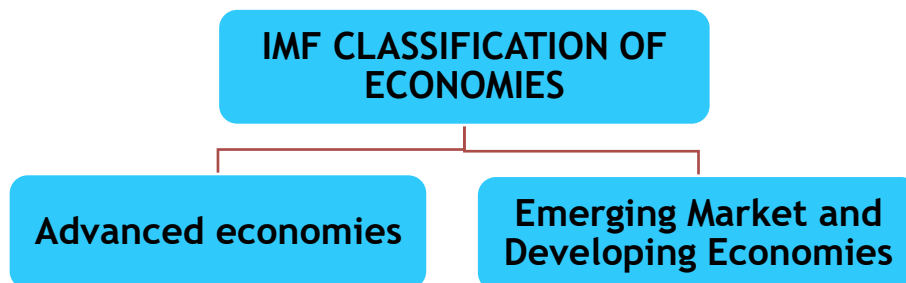
'National Income' and 'Per Capita Income'

	Low income	Lower-middle income	Upper-middle income	High income
July 1, 2025 (new, for FY26)	≤ 1,135	1,136 - 4,495	4,496 - 13,935	> 13,935
July 1, 2024 (previous, for FY25)	≤ 1,145	1,146 - 4,515	4,516 - 14,005	> 14,005



IMF's Classification

The economy classification in the WEO divides the world into two major groups:



- ❖ As per this classification India comes under Emerging market and Developing economies.

IMF uses following parameters for such classification:

- **Per capita income:** A country's per capita income level is a key criterion for classification.
- **Export diversification:** The extent to which a country diversifies its exports is another important factor.

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- **Global financial system integration:** How integrated a country is into the global financial system is also considered.

National Income Accounting in India

National Income of India is being measured by **Central Statistics Office (CSO)** since 1955 (currently merged with NSSO under NATIONAL STATISTICAL OFFICE), under The Ministry of Statistics and Programme Implementation.

The Central Statistics Office coordinates the statistical activities in the country and evolves statistical standards. It is headed by a Director General assisted by 5 Additional Director Generals. CSO has the following Divisions:

1). National Accounts Division (NAD): It is responsible for the preparation of **national accounts**, which includes Gross Domestic Product, Government and Private Final Consumption Expenditure, Fixed Capital Formation and other macro-economic aggregates.

2). Social Statistics Division (SSD): It is entrusted with **Statistical monitoring of the Millennium Development Goals, Environmental Economic Accounting, Grant-in-aid for research**, workshop/seminars/conferences in Official/Applied Statistics, National/International awards for Statisticians, National Data Bank (NDB) on socioreligious categories, Basic Statistics for Local Level Development (BSLLD) Pilot scheme.

3). Economic Statistics Division (ESD): This Division conducts **Economic Censuses and Annual Surveys of Industries (ASI)**, compiles **All India Index of Industrial Production (IIP)**, **Energy Statistics and Infrastructure Statistics**, and develops classifications like, National Industrial Classification (NIC) and National Product Classification (NPC).

4). Training Division: It is primarily **responsible for the training of manpower in theoretical and applied statistics to tackle the emerging challenges of data collection, collation analysis and dissemination required for evidence- based policy making as also for planning, monitoring and evaluation.**

The Division also looks after the National Statistical Systems Training Academy (NSSTA), which is a premier Institute fostering human resource development in official statistics in India as well as at international level, particularly amongst developing and SAARC countries.

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5). Coordination and Publications Division (CAP):

The Division looks after co-ordination work within CSO as well as within the Ministries and State/UT Governments in statistical matters, organizes Conference of Central and State Statistical Organizations (COCSSO) and ‘Statistics Day’(29th June) every year, prepares Results Framework Document (RFD), Citizens’/Clients’ Charter and Annual Action Plan, Outcome Budget and Annual Plan of the Ministry. **The administrative work relating to Indian Statistical Institute (ISI) is also looked after by CAP Division.**

National Sample Survey Office (NSSO) is doing tremendous work related to **primary data collection and analysis** through various NSSO rounds covering large number of socio- economic factors.

National Statistical Commission:

- ✓ **The National Statistical Commission, under the Chairpersonship of Dr. C. Rangarajan, had, *inter-alia*, recommended for the creation of National Statistical Office (NSO) to be headed by a National Statistician**, with appropriate autonomy and independence for producing Official Statistics as is prevalent in other countries.
- ✓ **The Government, while accepting the recommendations, had approved the establishment of NSO to be headed by Chief Statistician of India** by merging the then Central Statistical Organisation (CSO) and National Sample Survey Organisation (NSSO) to form the NSO in May, 2005.
- ✓ Since then, **The NSO was in existence** to serve as the executive wing of the Government in the field of statistics, and the Chief Statistician of India (CSI) & Secretary was its sole functionary.
- ✓ In the restructuring exercise, **the administrative, coordination and planning activities of MoSPI** have also been brought into NSO, which continues to be headed by CSI & Secretary, for providing requisite administrative support to the statistical activities of MoSPI.
- ✓ **NSC has five members including a Chairperson and four Members (seats are vacant). Besides these five members, there is a Secretary and an ex-officio Member in the NSC. The Chief Statistician of India is the Secretary of the Commission.** He is also the Secretary to the Government of India in the

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Ministry of Statistics and Programme Implementation. The CEO of the NITI Ayog is an ex-officio member of the NSC at present.

National Statistical Office (NSO):

- **Key Highlights:** The Government order has cleared restructuring of statistical system by proposing NSSO, CSO merger and formation of an overarching body called *National Statistical Office (NSO)*.
- **Objective of Restructuring:** to strengthen and streamline the present nodal functions of ministry and to bring in **more synergy** by **integrating its administrative functions within ministry**.

NSO Composition:

NSO to be headed by Secretary (S&PI), with various divisions reporting to Secretary through **the Director Generals (DGs)**.

There will be three Director Generals:

- 1) **DG (Statistics)**
- 2) **DG (Coordination, Administration and Policy)**
- 3) **DG (National Sample Survey) — reporting to the Secretary (S&PI).**